

Visualization of the Learning Process Using Concept Mapping

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Abstract

Visualization of the learning processes is a powerful way to help students to understand their curricula and the structure behind them. CME2 is a prototype software of this favourable concept, visualizing the learning process as a graph. With this tool students and teachers can easily handle and plan their personal curricula in larger scale and see the connections between different courses. The tool enables comparison between other students' curricula with set operations and it can be used as a planning tool for future studies. Within the teacher - student interaction this tool can be used as an effective aid for counselling.

1. Introduction

CME2 is a concept mapping [3] tool created in the Thinking Tools for the Net project at the University of Joensuu. The basic idea has been to create a tool for students and teachers to visualize and trace the learning process.

We started to create CME2 because there was not any concept map software capable to handle especially curricula. We wanted to create a tool for counselling that would be not just a repository for knowledge [4], but a tool to manipulate data.

Concept map technique was selected as a basis for this tool. Interestingly, we noticed that also Novak had presented a way to plan curriculum with concept map [5]; he stated that in curriculum planning a concept map can be an enormously useful [6] tool.

Since viewing data in a different way can help to gain additional insight into data, we saw lots of opportunities in this new approach. For example in the context of the virtual university this tool could be useful. Since these students are more responsible for their learning process than in the regular setting, an apparent need for tools like CME2 exists.

In our case, we define curriculum as a set of courses required for a degree. A curriculum consists of a set of individual courses that have some common relationship be-

tween them. This is a bit different from Novak's view; by curriculum he means the contents of an individual course [5].

Using concept maps for handling curricula significantly differs from traditional concept map uses. In CME2 we are not visualizing the domain knowledge [7] to be learned, but rather the process, or the way in which we are learning. This makes our approach different and opens new challenges for software developers in learning technology.

CME2 helps a student to get the data that concerns just him and is intended just for him. It makes the counselling more personal, even if the counsellor were not present.

2. Background

Virtual universities and the amount of courses given in digital environments are growing. This means that the need for counselling in the web is also increasing. Also noticeably in Finland, there is a need to shorten university students' studying times. All this development has forced faculties and departments to start projects to improve counselling [2], even at a larger scale. A good example of this development is the virtual study counselling project, OVI [2], run at the University of Joensuu.

Functional counselling methods are hard to implement [4] and several aspects should be considered. Contrary to existing systems, which are mostly passive repositories of instructional material [4], we should involve the student to take an active role in planning her studies. Future counselling systems have to activate a student [4] and force her to go deeper into counselling, to evaluate her studies, to find out the most appropriate learning paths for her. Because of its potential to bring the student to the best possible mental imagery of her knowledge [3], concept map is a good tool to represent formal knowledge, like curricula.

3. Technical design

Since the prototype serves as a curriculum planning and handling tool, some of its features differ from traditional

concept map editors. All these new features have been implemented to help the student create reasonable curricula and to ease work.

3.1. Concept map language

Since we deal with a novel use of concept maps, we defined a new language for storing them although several concept map languages can be found. Our language, called CoMaLa, is based on XML. When defining CoMaLa, the main intention was that it would be flexible, easy to understand and highly versatile. That is why only mandatory properties of map elements, like position for node, have been implemented in this language.

The CoMaLa language implements all the important properties for a concept map and its nodes and arcs. The main advantage of this language is that every concept map element can have as many attributes

3.2. Personal Counselling Tool

When connected to a database of available courses, CME2 can act as a personal instruction tool. Whenever a student adds a new course x to her personal curriculum, CME2 suggests other courses related to course x , to be added to the curriculum. With this feature, a student can build a logical and advisable curriculum quickly and observe all the time the structure of her studies.

With this operation CME2 can also work as a studyguide. Since all the courses are stored in database, CME2 can be used to explore the courses. When a student is exploring through the courses, he can see the dependencies between the courses and obtain the courses that might be useful for him.

3.3. Set Operations

It is often useful to compare different curricula with each other. For example, a tutoring teacher might want to know how her students meet the requirements in a specific area of expertise. Because extensive concept maps (like curricula) are quite hard to compare without tools, *set operations for concept maps* were implemented.

Set operations for concept maps include basic set operations, excluding complement. Union, intersection and difference operations can be defined and implemented rather easily for concept maps.

Each of these operations can also be presented as an animation. If the user wants to see what actually were the differences, or similarities between maps, animations are powerful. With an animation the user can see all the time how the process is proceeding and which objects are recognized to be similar. This approach helps him to really get insight into these two maps being compared.

3.4. Visualizing the Study Process

If CME2 is connected to student record database, reflecting the study process will become possible. This basically means that all the courses a student has passed can be marked, for example with a certain color. Also her grade, some other notifications and her own comments can be added to the map. This will help her, and her teacher, to analyze the study process.

For teachers who want to see how their students are doing, this allows quite an efficient way to do it. The teacher can easily see her students' progress, strengths and weaknesses and can help her in future studies.

4. Conclusion

We have presented a prototype of a concept map editor, called CME2. The software has features which make it particularly suitable for representing a dynamic study process, like tools for combining concept maps with set operations and adding descriptive information for the nodes. This is the main contribution of the present work in the area of concept mapping; earlier studies and projects emphasize features supporting static representations of a student's knowledge. The specific features of CME2 make it also a reasonable platform for supporting students' collaboration. In addition, concept map based process visualization could also be utilized in personnel management and training systems in business companies; adding a flavor of life-long learning in these contexts.

References

- [1] Gaines, B.R. and Shaw, M.L.G., Concept maps as hypermedia components, *Int. Journal of Human - Computer Studies* 43, p. 323-361, 1995.
- [2] Haapala, A., Virtual Study Counseling, OVI - project for developing virtual counseling and assessment environment. *Advancing Virtual University Education*, University of Joensuu, Finland, p. 152-163, 2001.
- [3] Kommers, P.A.M., Concept mapping, *Journal of Interactive Learning Research* 8, 3/4 p. 281-287, 1997.
- [4] Kuittinen, M., Pöntinen, S. and Sutinen, E., *How to Design Web-based Counseling Systems*, in Proc. ICAIT 2001, IEEE Learning technology task force, Madison, p. 178-179, 2001.
- [5] Novak, J.D. and Gowin, D.B., *Learning how to Learn*, Cambridge University Press, New York, 1984.
- [6] Novak, J.D., *The theory underlying concept maps and how to construct them*, <http://cmap.coginst.uwf.edu/info/> (Feb 5, 2003).
- [7] Rueda, U., Larraaga, M., Arruarte, M. and Elorriaga, J.A., *Using concept mapping tool for representing the domain knowledge*, in Proc. ICAIT 2002, IEEE Learning technology task force, Kazan, p. 387-391, 2002.